

Software Requirements Specifications (SRS) Document

Project: Home Network Solutions

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1. Introduction

1.1 Purpose

Home Networks Solutions will be a product that links many convenient and essential features across devices on a home network, as well as makes the network more comfortable to utilize in general. Seamlessly integrating features and services will allow the user to worry less about the linkage, or lack thereof, between devices. Instead, users can focus on their tasks at hand.

Current products available for home networking are technical in nature and require extensive server knowledge, which many clients do not have. Current solutions are technical in nature, requiring advanced knowledge. Solutions are also disjointed and may be incompatible with each other, resulting in many setup, maintenance, and troubleshooting issues.

Our product will make it simple enough to set up servers for non-technical clients whose workflow would benefit from the features of HNS. Our software will ensure security, stability, and ease of use fixing many of the issues current solutions contain.

1.2 Scope

Our software will have several key features such as file synchronization, network wide ad-block, IP sharing functionality, VPN support, AI agent support, and home security device support. The system will run on Ubuntu infrastructure and will support more devices capable of connecting to the home network such as mobile devices, desktops, smart TVs, etc. Users will be able to manage the system from their mobile devices or desktop remotely and securely.

Our system will use minimal pieces of hardware such as a Pi-hole for features such as ad-block. Our system will use a wide range of applications and open source projects to accomplish our goals. Our system will act as an all-in-one wrapper for the features provided by these tools.

1.3 Acronyms and Abbreviations

Term	Definition
HNS	The Home Network Solutions project
VPN	Virtual Private Network
IP	Internet Protocol
DNS	Domain Name System
AI	Artificial Intelligence

LLM	Large Language Model
UI	User Interface
OS	Operating System
LTS	Long-term Support
UI	User Interface
IoT	Internet of Things
Pi-Hole	Open source, DNS based network wide ad blocking application.
Tailscale	Mesh VPN service used to provide secure remote access to the local network.
Jellyfin	Open source media server used for video, photo, and audio streaming.
Home Assistant	Open source home automation platform used to manage IoT devices.
Caddy	Open source web server used to host web applications.
Samba	Open source implementation of the SMB protocol used for file sharing.
Jira	Project management tool used to organize and track the team's work.
Git	Version control system used to track changes to files.

1.4 References

- IEEE Recommended Practice for Software Requirements Specifications
- Tailscale Documentation: <https://tailscale.com/docs>
- Jellyfin Documentation: <https://jellyfin.org/docs/>
- Pi-hole Documentation: <https://docs.pi-hole.net/>
- Home Assistant Documentation: <https://www.home-assistant.io/docs/>
- Caddy Documentation: <https://caddyserver.com/docs/>
- Git Documentation: <https://git-scm.com/docs>
- Samba Documentation: <https://www.samba.org/samba/docs/>
- Jira Software Documentation: <https://support.atlassian.com/jira-software-cloud/resources/>

2. Description

2.1 Product Perspective

This product, by nature of streamlining several other programs, interacts with said programs frequently. This product will run in an Ubuntu Server environment and will largely interact with other programs using command line shell files.

The product will utilize Caddy to provide a local web server in order to interact with every program and feature using a simple and easy-to-understand graphical interface. The product will utilize Tailscale's tunneling software in order to enable the user to securely access the website from any network. The product will use Pi-hole and VPNs in order to provide an ad-block, as well as greater security, across the user's Tailscale network. The product will use Jellyfin to enable video and photo, and audio streaming across the user's network. The product will also use Home Assistant to enable common smart home features, such as managing smart devices and setting automations. Finally, Samba and Git will be utilized to enable file sharing and synchronization with version control across devices. All of these programs will be installed and set up during the product's setup.

2.1.1 System Interfaces

- Pi-hole
 - Provides network wide DNS based ad blocking as well as providing device connection data to the web application.
- Tailscale
 - Extends the home network to remote devices, allowing the server to be accessed outside of the home network.
- Home Assistant
 - Allows the web application to manage smart home devices.
- Jellyfin
 - Provides video, photo, and audio streaming.
- Samba
 - Enables file sharing between devices on the same network.
- Git
 - Version control and synchronization for user files.

2.1.2 User Interfaces

The primary user interface is a web application that consists of the following screens:

- Login screen
 - Seen on every visit containing a username and password field with error messages when the credentials are invalid.
- Setup wizard
 - Shown only on the initial setup. It walks the user through account creation and network configuration.
- Dashboard

- Has a collapsible side menu that links to the dashboard, applications, the file system, the network settings, and account/settings. In the middle, it shows a customizable widget area.
- Applications Page
 - An analytics page that shows installed programs.

For mobile devices, these pages will be shown in a layout suitable for smaller screens.

2.1.3 Hardware Interfaces

This product will be highly portable across hardware. The recommended device will be a Raspberry Pi 5, or a device with similar computing power. Weaker devices will be supported but may run into performance issues. The device must support Ubuntu Server 24.04 LTS and have network connectivity. The device should also have a wired network connection and expandable storage for reliability and longevity.

The product will use several ports. A majority of these ports will be dedicated to the website/web application, and the rest will be dedicated to various programs on the server. However, these ports will not obstruct other devices from forwarding their ports, as it will all be accessible locally through Tailscale.

2.1.4 Software Interfaces

Name	Version Number	Source
Ubuntu Server	24.04	https://ubuntu.com/server
Pi-hole	Latest	https://pi-hole.net
Tailscale	Latest	https://tailscale.com
Home Assistant	Latest	https://home-assistant.io
Jellyfin	12.1	https://jellyfin.org
Caddy	Latest	https://caddyserver.com
Samba	4.25.0	https://samba.org
Git	2.55.0	https://git-scm.com
SQLite	3.53.4	https://sqlite.org
Python	3.14.7	https://python.org

2.1.5 Communications Interfaces

- TCP/IP networking for communication between devices and the server.
- HTTP and HTTPS to deliver the web app and handle API requests.
- WireGuard VPN protocol for remote access via Tailscale.
- DNS queries for local network communication.
- SMB for file sharing across devices under Samba.

2.1.6 Memory Constraints

Due to the recommended hardware (Raspberry Pi 5 4GB), memory will be limited to 4GB. Storage will not be limited, as the storage size of the product will be rendered negligible compared to user-added items from usage, like videos and other media.

2.1.7 Operations

The user will not be required to perform any actions beyond the initial setup. However, by nature of the product, many features will only be useful to the user after configuration. Such features requiring user intervention include: uploading any desired media to the server, selecting folders or files to synchronize across devices, and adding and managing smart devices and their automations.

2.2 Product Functions

The key features and functionalities of the software are as follows:

1. File Synchronization System
 - i. The user will be able to access this system from any network.
 - ii. The user will be able to synchronize folders between different devices.
 - iii. If a file has changes made on multiple devices, the user will be able to choose which version to keep.
 - iv. The system will be secure and have essential cybersecurity features.
 - v. The user will be able to utilize version control akin to GitHub.
2. Ad-Block
 - i. The user will have a network-wide ad-block on all devices connected to the network.
 - ii. The ad-blocker will be configurable through the web application.
3. IP Sharing
 - i. The user will be able to share IP addresses between their devices.
 - ii. The user will have access to safety features and be able to configure who can and cannot use their IP address.
 - iii. The user will have access to the ability to configure this feature as they wish, including the ability to disable it.

4. VPN
 - i. The user will be able to run their home network's traffic through a VPN of their choice.
 - ii. The user will be able to swap seamlessly between VPNs.
 - iii. The user will be able to toggle on and off the VPN.
5. AI Agent
 - i. The user will be able to connect an AI agent of their choice, ie, local, Claude, Chat GPT, among others.
 - ii. The AI agent will be able to assist the user in configuring, diagnostic, etc, using inbuilt commands that the AI will have access to.
 - iii. The commands will be password protected and logged to prevent AI mistakes.
6. Home Security
 - i. Ability to connect home security devices to network, cameras, motion detectors.
 - ii. Ability to manage devices remotely.
7. Web Application
 - i. The user will be able to access all the features from a web browser.
 - ii. The web application will be available remotely.
 - iii. The system will be secure and have essential cybersecurity features.
 - iv. The application will be friendly to non-technical users.
8. Password and Passkey Manager
 - i. The user will be able to store personal passwords or passkeys to their home server.
 - ii. The user will be able to retrieve passwords or passkeys when needed.

2.3 User Characteristics

Users will be expected to have basic computer literacy skills, like connecting devices to their home internet or navigating an application's UI. Technical server knowledge will be helpful when using HNS software, but not necessary. No programming knowledge or deep software knowledge will be required. There will be optional, advanced configurations and settings for those who prefer it.

2.4 Constraints

1. Hardware should not need to be extremely powerful.
 - With the exception of running local LLMs.
2. The server must integrate with the third-party services, using each of their existing interfaces.
 - Third-party applications include Pi-hole, Tailscale, Home Assistant, Jellyfin, Caddy, Samba, and Git.

3. The server must comply with the licenses of each third party application.
4. The server must not lose or corrupt any user data in the event of power loss.
5. The server must not be compromised by the security failures of each third-party application

2.5 Assumptions and Dependencies

1. Hardware meets the requirements to run Ubuntu Server LTS.
2. The user has a home network capable of connecting to the Internet.
3. The user has devices capable of connecting to the home network.
4. The user's home network does not block connections used by HNS.
5. Third-party tools remain available for download and use.
6. The user is able to run the install script on the home server.

3. Specific Requirements

3.1 Functional Requirements

ID	Requirement	Description
FR1	Data storage	The system must be able to store and manage data of multiple types.
FR2	Connection Interception	The system must be able to intercept data streams that are being sent into it.
FR3	Networking	The system must be able to handle networking capabilities and handle incoming connections and disconnections.
FR4	Data Streaming	The system must facilitate data to stream through it from one device to another.
FR5	IP Sharing	The system must have the capability to share its IP address and or pass its data through a VPN provider.

3.2 Non-functional Requirements

ID	Requirement	Description
NFR1	Security	The system should have file encryption measures such as the secure folder. Will have a firewall to prevent unwanted connections. Shall have a ban list to ban specific connections and or data.
NFR2	Performance	The system should run on most modern hardware and services should not take longer than 30 seconds to load in optimal conditions, excluding network bottlenecks. File transfers should be proportional to the download speed of the network and size of the file, with the system adding minimal overhead.
NFR3	Usability	The system should have a user friendly UI with non technical verbiage to ensure easy usability. Extensive documentation of each tool, service, and features will be provided. AI agents will have access to diagnostic and configuration information which will be able to relay in a simplified manner to the user.
NFR3	Portability	The system should be based on an Ubuntu server. Smart devices should be able to connect to the system regardless of OS, such a phone, desktop, etc. The network will be accessible online allowing it to be accessed using mobile devices.
NFR4	Reliability	The system will have version control for files allowing for the recovery of corrupted and or accidentally deleted files. The system will have the ability to safely manage sudden disconnects of devices. The system will be able to handle a sudden outage of internet services. In case of a system shut down it will be able to restart safely without causing permanent damage.
NFR5	Maintainability	The system will be modular in nature allowing for the removal and addition of external features and applications to the server. The user will have access to diagnostic commands which will allow them to diagnose and fix issues as they come up. All instructions and source code will be shown with documentation.
NFR6	Reproducibility	The system should be reproducible on any home network.

3.3 Interface Requirements

- User interface: Web-based application
- Hardware requirement: Digital device able to connect and interact with a network
- Software interface: Ubuntu Server OS

3.4 Performance Requirements

The server will not take longer than 30 seconds to respond to a user request unless caused by an external factor (eg. Tailscale outage, poor internet connection). The server will be designed to handle one user at a time in order to minimize conflicts and allow more development time elsewhere, but it will likely be able to handle multiple concurrent users. Furthermore, data transfer speed should be limited so that multiple tasks can run concurrently without hogging resources.

3.5 Security Requirements

Security is very important due to the nature of the data that will be stored on this home server. As such, no unauthorized devices should be able to access the server. This will be achieved through only allowing devices to connect through Tailscale or the local network. Furthermore, after connection, the user must log into their account to be able to access the server, adding a second layer of security. Finally, all files will be stored locally, which, in combination with the restrictions on which devices can connect, will keep the files safe from anyone without an authorized device and login.

3.6 Design Constraints

The HNS software will be designed to run on low-power hardware, notably a Raspberry Pi 5. This specific model has a quad-core ARM CPU with a 2.4 GHz clock, and 4 GB of LPDDR4 memory. As such, computing resources will be limited. The requirements for Ubuntu Server recommend at least 1.5 GB of memory, with 3 GB recommended. These requirements consume a large portion of the hardware's resources, leaving little headroom for extra processes or services.

Since the hardware will be employed as a server, the strength and bandwidth of its link to the home network is critical. The hardware should be hardwired to an internet router via an ethernet cable, but there may be instances where this is not possible. So, we must keep in mind that users may prefer the server connection over WiFi, which could introduce latency and data integrity issues. There may also be multiple services communicating with multiple devices at once, so the resources consumed by each connection must be minimal.

3.7 System Attributes

- Ubuntu Linux Version: Ubuntu Server LTS 24.04
- Tailscale: Latest Version
- Pi-Hole: Latest Version

- Caddy: Latest Version
- Jellyfin: Latest Version
- Home Assistant: Latest Version
- Git
- Samba